

Charan Lingamgunta

STAFF SOFTWARE ENGINEER

AI PLATFORM | AGENT INFRASTRUCTURE | DOCUMENT AI | LLM EVALUATION | DISTRIBUTED SYSTEMS
Boston, MA | charan.lingamgunta@gmail.com | linkedin.com/in/charan-lingamgunta | charan-lingamgunta.vercel.app

PROFESSIONAL SUMMARY

Staff Software Engineer with 15 years in backend and distributed systems for regulated healthcare, financial services, and public-sector customers, the last 5 in applied AI. Architect of SecureGPT, building the infrastructure its AI products ship on, the agent harness, document AI pipeline, model gateway, guardrails, evaluation, and model serving, and taking each from architecture to production in customers' own environments. Technical lead for 12 engineers across four AI teams. Earlier, built an enterprise API gateway carrying 10B+ transactions a day.

CORE SKILLS

Agents & Document AI: Agent Infrastructure (harness, runtime, tool use, MCP), Sandboxed Code Execution, Document AI & Extraction (OCR, table layout, vision-language), Hybrid RAG (BM25 + kNN, RRF), Evals & Quality Gates (LLM-as-Judge, golden datasets, A/B), Safety Guardrails & PII De-identification, AI Gateway & Model Routing (LiteLLM), Prompt Optimization (DSPy)

LLM Serving & Training: vLLM inference (multi-GPU tensor/expert parallel, speculative decoding), Fine-Tuning (PEFT, LoRA), MLflow (GenAI eval, tracing, observability), OpenTelemetry, PyTorch, CUDA, Ray, Hugging Face

Distributed Systems: Event-Driven Architecture, Streaming APIs (SSE), Pub/Sub & Messaging (JMS, AMQP, Kafka), API Gateways, Multi-Tenant Systems, Reliability & High-Availability Design

Cloud & Tooling: AWS (EC2, S3, RDS, ECS, Lambda, Bedrock), GCP, Docker, OpenSearch, Redis, Airflow, SkyPilot, GitHub Actions

Languages: Python, Java, TypeScript, JavaScript, C#

Frameworks & Runtimes: FastAPI, Django, React, Node.js

Identity, Security & Compliance: OAuth 2.0, SAML, LDAP, HSM, FIPS 140-2; HIPAA, HITRUST, PCI-DSS, SOC 2

PROFESSIONAL EXPERIENCE

FORUM SYSTEMS (QUANTUM GEARS), Needham, MA

Staff Software Engineer	Jan 2022 - Present
Senior Software Engineer	Sep 2016 - Dec 2021
Software Engineer	Jun 2014 - Aug 2016

MIMECAST, Watertown, MA

Software Developer Intern	Jun 2013 - Aug 2013
---------------------------	---------------------

APPLABS (CSC), Hyderabad, India

Associate Software Engineer	Dec 2009 - Jun 2012
-----------------------------	---------------------

PROJECTS

SecureGPT, Document AI and Agent Harness, Forum Systems (Quantum Gears) Jan 2022 - Present

- Built the chat and document chat inference service SecureGPT launched with, private and public models behind guardrails and de-identification, streaming responses to users.
- Lead the SecureGPT AI stack. Set the architecture, evaluation, and security standards that 12 engineers across the inference, agents, document ingestion and RAG, and deployment teams build to, and chair architecture reviews.
- Built the agent harness behind 10,000+ governed agent runs a day for analyst and data science teams at a top-3 health insurer. It orchestrates nested sub-agents as tools, streams results live or runs durably in the background with checkpoint and resume, and traces every run in MLflow.
- Designed the document ingestion engine. Uploads take one of three extraction paths, fast text, OCR with table layout, or vision-language (VL) deep transformation, then land in OpenSearch with self-hosted embeddings and PII and PHI tags.
- Raised fact-level extraction accuracy from 63.3% to 97% on a 685-fact ground truth, for the pipeline reading 50,000 insurance and Medicare documents a month, by reworking the extraction pipeline and benchmarking frontier against open-weight VL models to pick the production model.
- Established agent evaluation with golden datasets as regression gates before release, and online evals that sample production runs, score them with an LLM judge, and flag failures for review.
- Shipped the guardrails, five concurrent checks on every prompt (legal-risk screening, acceptable-use, content moderation, intent classification, and safety) plus prompt-injection detection, extensible with a firm's own guardrail

prompts. Paired them with de-identification that replaces names, addresses, and health identifiers before a public model call and restores them in the answer.

- Engineered the sandbox the harness runs agents and tools in, a restricted Python interpreter that enforces AST validation, import allowlists, and resource limits from sandbox testing through production, with optional per-agent microVM isolation.
- Extended the agent harness with an MCP server and tool registry that scopes discovery to the namespaces a query needs, with per-identity authorization and SOC 2-aligned audit logging.
- Architected the agentic RAG pipeline on OpenSearch hybrid retrieval, BM25 + kNN with reciprocal rank fusion and query decomposition.
- Cut peak memory on document load by 70% by instrumenting the embedding pipeline and calibrating chunk sizes to its memory profile, which unblocked parallel document processing.
- Own the model serving layer, bringing up open-weight models on vLLM across AWS, GCP, and RunPod GPU hosts and frontier models through Bedrock, all behind one OpenAI-compatible LiteLLM gateway with access policies.
- Built the model evaluation harness that scores each frontier and open-weight release on agentic reasoning, code generation, PDF extraction, guardrail, and de-identification tasks against internal ground truth, and benchmarks open-weight models for throughput across GPU topologies. Promising models are A/B tested and phased into production.

LLM Fine-Tuning, Forum Systems (Quantum Gears)

Jan 2023 - Jun 2025

- Built the fine-tuning platform behind 20+ production models at enterprise firms, with PEFT/LoRA training on versioned datasets and reproducible splits, checkpointed training with pause and resume tracked in MLflow, and release after testing and approval.
- Reduced training cost with SkyPilot spot provisioning and Ray distributed training across AWS and GCP; served the models on vLLM behind the gateway.

DevX, API Catalog, Agent & Tool Authoring, Forum Systems (Quantum Gears)

Jan 2022 - Present

- Lead architecture for the multi-tenant platform where a firm catalogs its APIs, authors agents and tools, and turns API operations into MCP-served agent tools through its SDK, published into the SecureGPT harness. The largest deployment covers 200+ engineers and 1,000+ APIs exposed as 2,500+ agent tools at one Fortune 500 firm.
- Built the self-serve authoring flow analysts and data scientists use to write, sandbox-test, and publish their own agents and tools.
- Enforced role-based access control (RBAC) on published agents and tools and attribute-based access control (ABAC) on data access and egress at runtime.

Forum Sentry, Enterprise API Gateway and Builder, Forum Systems (Quantum Gears)

Jun 2014 - Dec 2021

- Architected the 4-stage bidirectional streaming pipeline carrying 10B+ transactions daily in aggregate across customer deployments, over HTTP, SOAP, JMS, AMQP, SFTP, and S3, with zero-copy passthrough, automatic retry and failover, and stage-level latency attribution.
- Cut backend load 40-70% with multi-tier API response caching (memory, disk, Redis) and request coalescing for cache-stampede protection, configured through no-code cache policies.
- Designed the no-code API builder with dozens of processors for security, data transformation, routing, and validation, sandboxed inline JavaScript, and other integrations that cut new-API onboarding from weeks to hours.
- Engineered active-passive failover with sub-minute recovery and 99.99% uptime for financial and government workloads on FIPS 140-2, HSM-backed deployments.
- Built the authentication subsystem supporting 11 authentication methods; its caching and connection pooling eliminated 90%+ of directory roundtrips.

MimeDrive, Mimecast

Jun 2013 - Aug 2013

- Enterprise email encryption application in C#/WPF with PowerShell test automation.

Blackboard Learn, AppLabs (CSC)

Dec 2009 - Jun 2012

- Reporting and notification modules for an e-learning platform serving millions of users.

EDUCATION

Master of Science in Computer Information Systems, Northeastern University, Boston, MA

2012 - 2014

Bachelor of Engineering in Computer Science, JNTU, India

2005 - 2009